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LETTER TO THE EDITOR

A decision algorithm for assessing indirectness in core GRADE

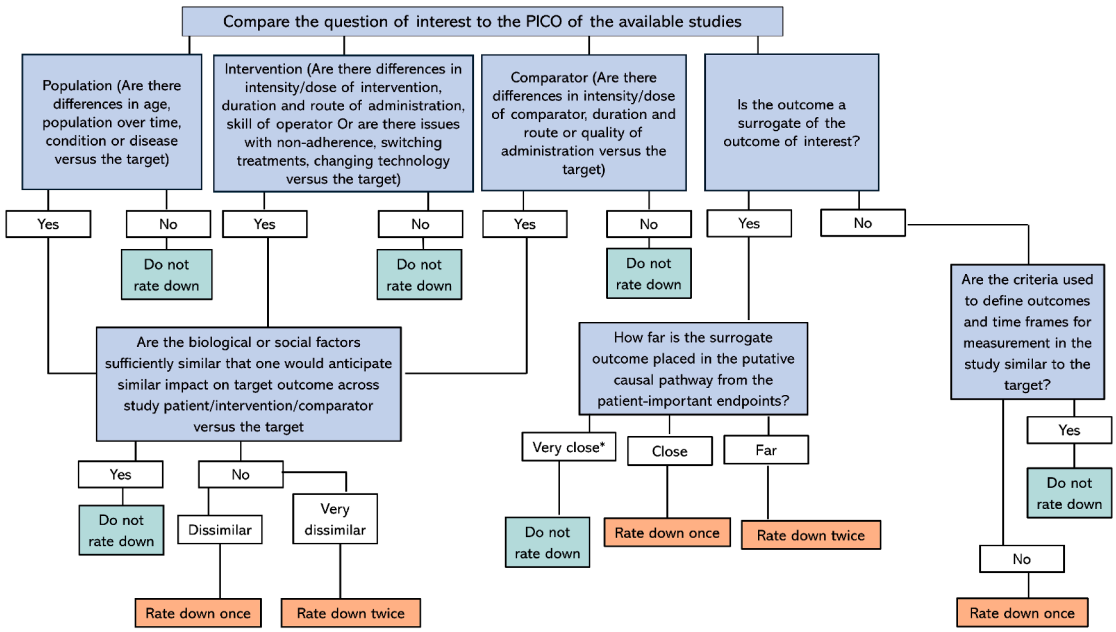
Letter to Editor,

In the recent Core GRADE (Grading of Recommendations Assessment, Development and Evaluation) series, article 5—*Rating certainty of evidence—assessing indirectness* [1], published in The British Medical Journal, and in the complementary Journal of Clinical Epidemiology article, *Core GRADE Unpacked: A Summary of Recent Innovations in Complementary GRADE Methodology* [2], we outlined the Core GRADE principles and explained how Core GRADE users can judge whether or not to rate down for indirectness when Population, Intervention, Comparator, or Outcome (PICO) elements diverge from the target PICO. Subsequent to these publications, we have reflected on the desirability of an algorithm to assist users in making this decision and decided that such an algorithm would be helpful. Figure presents the algorithm as a logical, stepwise complement to the existing text.

The process begins by determining whether the evidence is direct or indirect. Although indirectness is rarely a concern in systematic reviews of direct evidence, it is frequently an issue in guidelines or HTAs in which the PICO in the relevant studies differs from the target PICO. Thus, when recognition of the limitations of direct evidence (low or very low certainty for a particular outcome) prompts an explicit search for indirect evidence, serious concerns about indirectness are very likely to arise.

For Population-Intervention-Comparator, the algorithm first prompts assessors to judge whether there are differences—in, for example, age, disease severity, intervention dose, clinician expertise, or comparators—between the target and study PICOs. If there are differences, it asks whether biological or social factors are sufficiently similar to expect the same impact on the target outcome. The degree of dissimilarity then dictates whether to rate down once, twice, or not at all.

For outcome, the algorithm focuses on two common sources of indirectness: surrogate outcomes and mismatches in outcome definition or timing. It directs assessors



*Although some outcomes may be considered surrogates, they are so closely and directly linked to the disease process that they always represent a patient-important outcome. For example, a hip fracture diagnosed on radiograph inevitably leads to pain, immobility, and the need for treatment, and in malaria, treatment failure (recurrence or recrudescence) confirmed by uncorrected or corrected PCR is a patient-important outcome.

Figure. Decision-algorithm for assessing indirectness in core GRADE.

[1] Both authors contributed equally (joint first authors).

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to judge how closely a surrogate links to the patient-important endpoint and whether the measurement aligns with what matters to patients [3].

We anticipate that this visual decision-support tool will improve consistency and transparency in indirectness assessments, enhance GRADE training, and serve as a reference for systematic reviewers, health technology assessors, and guideline developers.

CRediT authorship contribution statement

Gordon Guyatt: Conceptualization, Visualization, Writing − review & editing. **Prashanti Eachempati:** Conceptualization, Methodology, Visualization, Writing − original draft. **Monica Hultcrantz:** Validation, Writing − review & editing. **Victor Montori:** Methodology, Validation. **Per Olav Vandvik:** Methodology, Validation, Writing − review & editing. **Alfonso Iorio:** Methodology, Validation. **Thomas Agoritsas:** Methodology, Supervision, Validation, Writing − review & editing.

Declaration of competing interest

The authors declare that they have no conflicts of interest in relation to this submission. No financial, personal, or institutional relationships have influenced the development of this algorithm or the preparation of the accompanying letter to the editor.

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Data availability

No data was used for the research described in the article.

References

[1] Guyatt G, Iorio A, De Beer H, Owen A, Agoritsas T, Murad MH, et al. Core GRADE 5: rating certainty of evidence—assessing indirectness. BMJ 2025;389:e083865. https://doi.org/10.1136/bmj-2024-083865.

[2] Colunga-Lozano LE, Wang Y, Agoritsas T, Hultcrantz M, Iorio A, Montori VM, et al. Core GRADE unpacked: a summary of recent innovations in complementary GRADE Methodology. J Clin Epidemiol 2025;189:112047. https://doi.org/10.1016/j.jclinepi.2025.112047.

[3] Christensen R, Ciani O, Manyara AM, Taylor RS. Surrogate endpoints: a key concept in clinical epidemiology. J Clin Epidemiol 2024;167:111242. https://doi.org/10.1016/j.jclinepi.2023.111242.

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